

CPW-fed Hexagonal Modified Sierpinski Carpet Fractal Antenna for UWB Applications

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Abstract— In this paper, a coplanar waveguide (CPW)-fed planar fractal antenna for bandwidth enhancement is designed. The proposed antenna consists of two iterations modified Sierpinski square slots with hexagonal radiation patch and CPW feed line. The simulated results indicate an impedance bandwidth of 133.5% (over 2.86-14.36 GHz) and a maximum peak gain of 5.5 dBi with stable radiation pattern. This antenna is designed on Rogers RO4350B substrate with height of 1.524 mm and 26×25 mm² of size. The designed antenna has a good potential to be used for UWB applications with compact size and better performance.

Keywords—Coplanar waveguide feedline; Sierpinski Fractal; Hexagonal patch; UWB applications.

I. INTRODUCTION

In recent years, the field of communication has made enormous progress in meeting the growing needs of users, which will require not only the development of successful communication systems, but also their adaptation to specific needs [1].

The Ultra-Wideband technique is one of the emerging technologies that has attracted a lot of attention of many researchers [2]. This technology has several advantages, such as high capacity and more immunity to multi-path cancellation [3]. Recently, planar fractal monopole antennas have been introduced as one of the major methods to integrate the UWB systems [4]. Furthermore, fractal geometry has properties of space filling and self-similarity leading to antenna size reduction, multiband and broadband characteristics [5].

In this paper, a Hexagonal UWB based on fractal geometry is proposed and investigated. This antenna is composed of a modified Sierpinski square slots-cut hexagonal patch monopole which is mounted on a dielectric layer with a CPW-fed transmission line. The simulated reflection coefficient S_{11} and radiation patterns were extracted from FIT-based on electromagnetic simulation CST MWS software [6].

II. DESIGN OF THE PROPOSED ANTENNA

The proposed antenna is designed on 26×25 mm² Rogers RO4350B substrate material with a relative permittivity of $\epsilon_r=3.66$, a thickness of $h=1.524$ mm, and loss tangent $\tan \delta=0.004$. The antenna is fed by coplanar waveguide (CPW) line, with a gap distance $g=0.3$ mm, and width $W_f=3.6$ mm. The layout evolution of the Hexagonal Sierpinski Carpet antenna for UWB applications is depicted in Fig. 1. Three steps have been carried out to extend the impedance bandwidth of the antenna over the entire UWB.

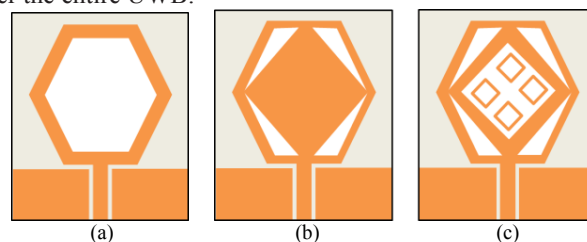
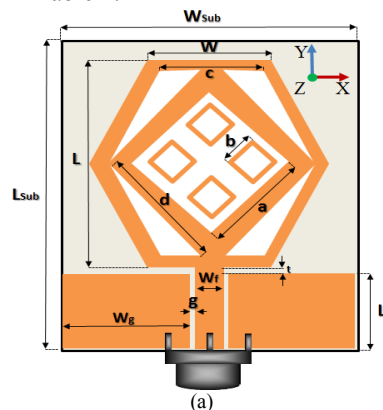


Fig 1. Iterative structures of the Hexagonal Modified Sierpinski Carpet fractal antenna: (a) basic hexagonal form, (b) First iteration and (c) Second iteration.

The optimized configuration of the proposed fractal antenna is shown in Fig. 2 and the geometrical parameters are summarized in Table 1.



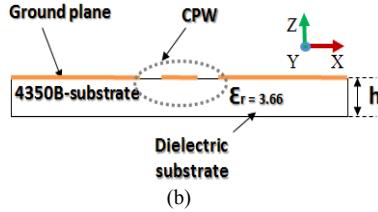


Fig 2. Geometry and configuration of the designed antenna (units in millimeter): (a) Top view, (b) Side view.

TABLE I. OPTIMISED DIMENSIONS OF THE PROPOSED ANTENNA.

Parameter	Value (mm)	Parameter	Value (mm)
a	9.8	<i>W_{sub}</i>	25
b	2.8	<i>L_{sub}</i>	26
c	8.3	<i>L_g</i>	6.25
d	9.3	<i>W_f</i>	3.6
h	1.524	<i>W_g</i>	10.4
g	0.3	<i>t</i>	0.9
W	10.5	<i>L</i>	18.2

III. RESULTS AND DISCUSSION

A. Reflection Coefficient S_{11}

The simulated S parameter results of the modified Sierpinski antenna for different iterations are illustrated in Fig. 3. From these results, it can be observed that the impedance bandwidth of the proposed antenna has an UWB bandwidth characteristics from 2.86 GHz to 14.36 GHz ($S_{11} \leq -10$ dB). Furthermore, it can be noted that the impedance bandwidth is improved with increasing of the iteration number of the proposed antenna.

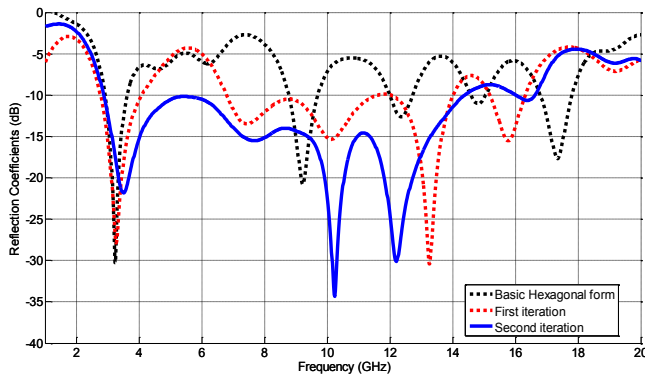


Fig 3. Simulated reflection coefficients S_{11} versus frequency with and without fractal elements.

B. Radiation Pattern and gain

Fig. 4 shows the simulated radiation pattern characteristics of the proposed UWB antenna at three peak frequencies 3.5 GHz, 7.5 GHz and 10.2 GHz, respectively. It can be seen that the simulated patterns are almost Omni-directional radiation pattern behavior in XZ plane and symmetric in YZ-plane at the operating frequencies. It is worth mentioning that the realized gain of Fig. 5 increases with frequency (from 0.5 to 5.5 dBi).

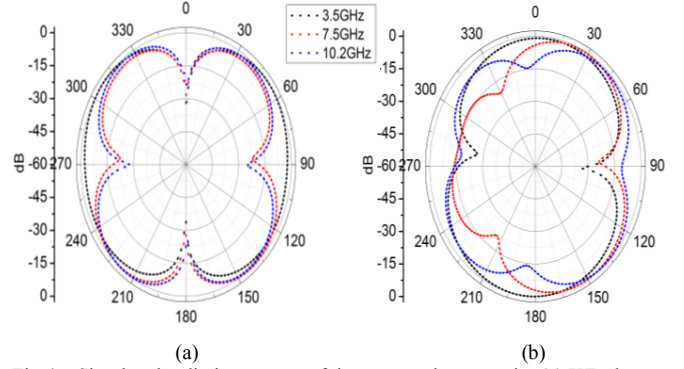


Fig 4. Simulated radiation pattern of the proposed antenna in: (a) XZ-plane, (b) YZ-plane.

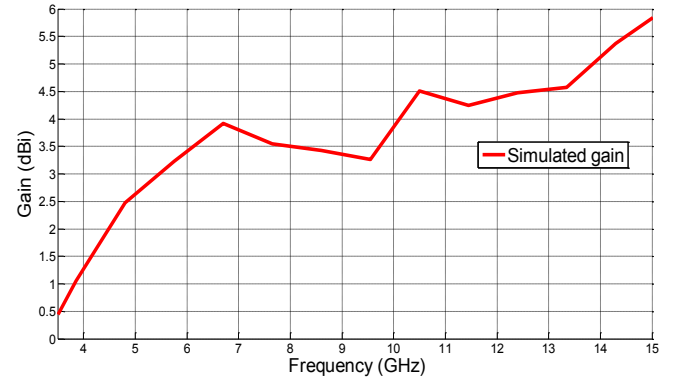


Fig 5. Simulated peak gain versus frequency.

IV. CONCLUSION

A Hexagonal modified Sierpinski carpet antenna with CPW-Fed has been presented. The obtained results showed that the proposed antenna was able to achieve an impedance bandwidth from 2.86 GHz to 14.36 GHz. The proposed antenna provides a good performance in terms of compactness and simplicity. With this feature, the antenna is suitable for ultra-wideband communications.

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